

SEEDPRINTS: FINGERPRINTS CAN EVEN TELL WHICH SEED YOUR LARGE LANGUAGE MODEL WAS TRAINED FROM

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ABSTRACT

Fingerprinting of Large Language Models (LLMs) is a prerequisite for provenance verification and model attribution. Prevailing fingerprinting methods are evaluated almost exclusively in the post-fine-tuning regime, in which a model has already accumulated stable signatures from its training corpus, its optimization trajectory, or its hyperparameter configuration. Because the majority of a model’s representational capacity and encoded knowledge is acquired during pretraining rather than during downstream fine-tuning, large-scale pretraining constitutes the more fundamental regime in which lineage must be verified. We demonstrate that existing fingerprinting methods are *unreliable* in this regime: their signatures are defined post hoc and emerge only after substantial optimization, a property that violates the classical Galtonian criterion of a fingerprint as an intrinsic and temporally invariant identity. We instead formalize a stronger and more intrinsic notion of LLM fingerprinting and instantiate it as **SeedPrints**, a procedure that exploits the biases induced by random parameter initialization as persistent, seed-dependent identifiers that are present prior to any gradient update. Concretely, we establish that untrained models exhibit reproducible, seed-determined prediction biases, and that these weak signals remain statistically detectable throughout optimization, thereby supporting high-confidence lineage verification. Whereas prior techniques fail during early pretraining or degrade under distribution shift, **SeedPrints** retains its discriminative power across the entire training lifecycle, spanning initialization, large-scale pretraining, and downstream adaptation. Controlled experiments on LLaMA-style and Qwen-style architectures establish seed-level distinguishability and enable birth-to-lifecycle identity verification. Further evaluations on large-scale pretraining trajectories and real-world fingerprinting benchmarks confirm robustness under prolonged optimization, domain shift, and parameter-altering transformations.

Taken together, these results establish that initialization alone imprints a unique and persistent identity on an LLM, yielding a fingerprint that satisfies the “Galtonian” criterion. Code will be released upon acceptance.

1 INTRODUCTION

LLM fingerprints have recently been introduced as a mechanism for identifying, attributing, and tracing LLMs on the basis of their observable behaviors (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024). The objective of such methods is to furnish a model owner with a verifiable link between a suspect model and its putative source, thereby enabling the detection of model theft or unauthorized reuse (Yoon et al., 2025; Zhang, 2025).

A substantial portion of this literature explicitly appropriates the metaphor of biological fingerprints from Francis **Galton**’s *Finger Prints* (1892) (Galton, 1892):

“A fingerprint is the pattern formed by friction-ridge skin on the fingertips; this ridge configuration is individually unique and essentially permanent across an individual’s lifetime.”

SeedPrints vs Baselines: OLMo-2-7B Pre-training Checkpoints

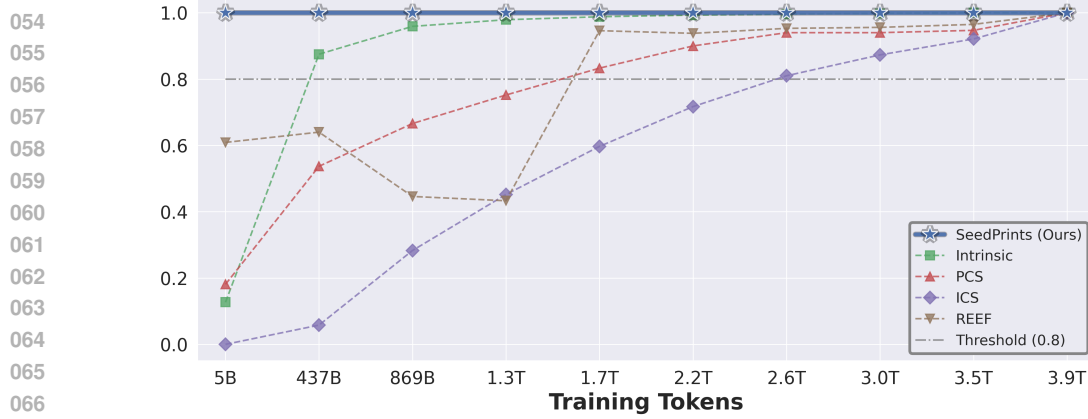


Figure 1: **Existing fingerprinting methods fail to detect model lineage during early pretraining.** We evaluate five methods on OLMo-2-7B checkpoints spanning 5B to 3.9T training tokens, each tested against the final checkpoint as the reference. The vertical axis reports the similarity score, for which larger values indicate a stronger lineage signal; the dashed horizontal line denotes the 0.8 detection threshold. All baselines degrade monotonically and cross below the threshold at the early checkpoints, whereas **SeedPrints** attains perfect detection ($p \ll 0.001$, reported as $1 - p$ for directional consistency with the baselines) from the first checkpoint onward.

This analogy imposes two formal requirements on an effective fingerprint: it must be *unique* across identities and *persistent* from the instant of a model’s “birth” at initialization. Most existing fingerprinting approaches, however, satisfy neither requirement (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024; Zhang, 2025; Luan et al., 2025; Tsai et al., 2025; Alhazbi et al., 2025). As reported in Figure 1, these methods do not reliably detect lineage during early pretraining. The underlying cause is that such signatures are defined only once a model has substantially converged, e.g., by extracting statistical patterns from parameters or from generated text. Consequently, the separability they exhibit reflects not an intrinsic birthmark of the model but rather an imprint of surface-level training factors, including data signatures, optimization dynamics, and hyperparameter choices. As we show in Section 5.1, existing baselines frequently misattribute lineage once the training data distribution shifts substantially. Such methods therefore operate as post hoc identifiers rather than as **Galtonian fingerprints**—the innate, immutable marks that accompany a model from its inception.

In this work, we formalize a stricter notion of LLM fingerprinting: an intrinsic property that is instantiated at *initialization* and remains detectable at *any* subsequent point of the training trajectory. Our principal contributions are as follows.

- **Characterizing a fundamental limitation of existing methods.** We establish that prior fingerprinting approaches fail to reliably detect lineage during early pretraining and are readily misled by substantial shifts in the training data distribution. (Section 5.1)
- **Identifying and exploiting initialization-driven fingerprints.** We show that untrained models exhibit seed-dependent biases in their internal representations, constituting a weak but persistent identity signal that is present from initialization. (Section 3) On this basis we introduce **SeedPrints**, a procedure that isolates these initialization-born fingerprints and exhibits improved robustness to confounding factors including data distribution, training duration, and optimization dynamics. (Section 4)
- **Comprehensive empirical validation.** We demonstrate that the proposed method separates models that differ only in initialization seed, even under identical training pipelines and fixed data orderings, and that this separation persists under continual training across heterogeneous datasets. (Section 5.1)
- **Strong performance under realistic conditions.** The method reliably detects lineage throughout large-scale pretraining trajectories (e.g., OLMo-2-7B Stage 1) without early-stage failure. It further attains strong performance under large-scale fine-tuning (with data volumes up to 700B tokens) and in practical deployment scenarios drawn from the *LeaFBench* benchmark (Shao et al., 2025), which comprises 65 model instances spanning 7 mainstream model families and 6 transformation classes (instruction tuning, fine-tuning, PEFT, quantization, model merging, and distillation). In

these settings the method matches the strongest baseline (near-perfect detection) while substantially outperforming all remaining baselines (Sections 5.2 and 5.3).

2 RELATED WORK

Approaches to LLM protection partition into two families: (i) *watermarking / active fingerprinting* methods that *insert* an identifiable signature into a model or its outputs (Xu et al., 2024; Nasery et al.; Tsai et al., 2025; Sander et al., 2024), and (ii) *passive fingerprinting* methods that *extract* a signature from a model’s pre-existing behaviors without any modification (Yoon et al., 2025; Zhang et al., 2024; Alhazbi et al., 2025; Pasquini et al., 2024; Suzuki et al., 2025; Zhang, 2025).

Watermarking and Active Fingerprinting. Active approaches deliberately implant a verifiable identifier to support later ownership verification. Representative techniques include backdoor insertion (Adi et al., 2018; Li et al., 2019; Zhang et al., 2018) and digital signatures or hash functions (Guo & Potkonjak, 2018; Zhu et al., 2020). For language models, two instantiations predominate: *text watermarks*, which bias the generation distribution or insert predefined token patterns to encode hidden information (Kirchenbauer et al., 2023; Xu et al., 2024; Nasery et al.); and *model-weight watermarks*, which embed identifiers directly into parameters or bind them to secret triggers via fine-tuning (Luan et al., 2025; Li et al., 2023). Although backdoor-style fingerprints are conceptually simple and can persist through moderate fine-tuning (e.g., Dasgupta et al. 2024), these invasive schemes presuppose control over the training process, which renders them inapplicable to the retroactive marking of third-party models.

Passive LLM Fingerprinting. In contrast, passive fingerprinting identifies a model by analyzing its intrinsic properties without any modification, and the applicable techniques are stratified by the level of model access. Under **white-box access**, the signature is extracted from the model weights, exploiting intrinsic quantities such as the distribution of attention matrices (Yoon et al., 2025), the kernel alignment of internal representations (Zhang et al., 2024), or the stable directions of parameter vectors. Under **black-box access**, the signature is derived from input–output behavior: such methods rely on crafted queries (Pasquini et al., 2024), unique prompt–response pairs (Tsai et al., 2025), stylometric features (Alhazbi et al., 2025), or iterative prompting–response games (Iourovitski et al., 2024) to identify a model, at the cost of reduced robustness to fine-tuning. Critically, all of these methods define their signatures *post hoc*: they extract properties that emerge from a completed training process rather than the innate, “Galtonian” fingerprints present at random initialization that this work targets. As a direct consequence, *they are not designed for lineage verification during large-scale pretraining*, the regime in which such post-training signals have not yet emerged.

3 BIASES ORIGINATING FROM INITIALIZATION PERSIST AFTER TRAINING

In this section we present our central empirical observation: language models exhibit pronounced prediction biases that originate from the random initialization seed, and these biases remain statistically detectable after training.

Initialization induces seed-specific bias profiles. We instantiate a LLaMA-2–style model with seed 123 and evaluate it on 10,000 random input sequences of length 1,024, in which token identifiers are drawn i.i.d. and uniformly from the vocabulary. In Figure 2 (Left), we plot, for each output token (upper panel, computed from the final logits) and each hidden dimension (lower panel, computed from the last-layer representations), the empirical frequency with which that coordinate attains the minimum value across the 10,000 random trials. The pronounced deviation from uniformity in both panels evidences the existence of extreme output-preference bias patterns. Repeating the procedure across distinct initialization seeds reveals that, although the overall magnitude of the bias is approximately invariant, the specific argmin coordinates are seed-dependent (Figure 2, Lower Right). These observations are consistent with recent analyses attributing the bias of randomly initialized models to inter-sequence representation contraction, which is induced by asymmetric nonlinear activations in the MLP blocks and subsequently amplified by self-attention (Li et al., 2026). Within this account, distinct initialization seeds induce distinct contraction directions, which in turn yield distinct argmin patterns over the output coordinates.

Training preserves initialization-born bias. Although optimization substantially alters the magnitudes of the outputs, the relative preference ordering over output coordinates that is induced at initialization is not fully erased. We continue training the initialized model (seed 123) for one epoch

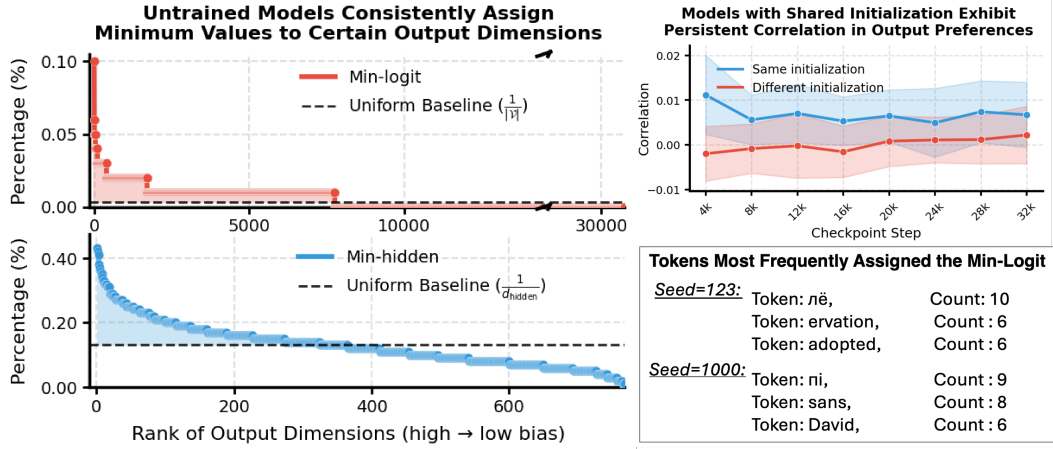


Figure 2: **Initialization-born output bias persists through training.** *Left:* Under fully random inputs, the outputs of a randomly initialized LLaMA-2-style model deviate markedly from uniformity and instead exhibit a systematic bias, whereby certain coordinates are disfavored (i.e., they attain the minimum value across random inputs with high frequency). This extreme bias is present both in the logits (top, red) and in the final hidden representations (bottom, blue). The dashed line marks the expected frequency under a uniform distribution, and the arrows in the top panel denote a broken x -axis that omits the low-frequency tail ranks. *Upper Right:* Throughout training, each model retains a weak positive correlation, computed across inputs, with the base model from which it was trained. This correlation distribution is consistently offset upward relative to correlations computed against independently initialized base models, indicating that initialization-born bias leaves a measurable signal that survives training. *Lower Right:* The coordinates exhibiting the strongest bias are determined by the initialization seed; we illustrate this using the tokens that most frequently receive the minimum logit.

on OpenWebText (Gokaslan et al., 2019) and evaluate its intermediate checkpoints. We restrict attention to the $m = 50$ output coordinates that are most frequently assigned the minimum value by the initialized model. For each checkpoint, we compute the correlation, over random inputs and restricted to these coordinates, between the checkpoint’s responses and those of the initialized model; this quantifies the degree to which the checkpoint and the initialized model rank or disfavor the selected coordinates consistently across inputs. As a reference, we compute the identical correlation against an independently initialized model (seed 1000). As shown in Figure 2 (Upper Right), although the absolute correlation values are small, the correlations for models sharing an initialization are consistently offset upward relative to the independent reference and stabilize rather than decaying to zero. Collectively, these results establish that initialization deposits a weak but statistically detectable bias signal that persists throughout training.

4 ALGORITHM

Section 3 established that models belonging to the same initialization lineage retain a weak but consistent agreement over a subset of output coordinates (specifically, those most disfavored by the untrained model under random inputs). Our objective is therefore twofold: to recover, for any pair of models and without access to the underlying initialized model, a set of coordinates that encode this bias signal, and to test whether the models’ agreement over that set is statistically significant. The complete procedure is summarized in Algorithm 1; we specify each component below.

Extracting identity dimensions between any two models. Let $X = \{x_i\}_{i=1}^n$, where each $x_i \in \mathbb{R}^{\ell \times d}$ is a random sequence of length ℓ . Each input x_i is instantiated in one of two ways: by sampling ℓ token identifiers uniformly from the vocabulary, or by sampling ℓ independent vectors from a d -dimensional isotropic Gaussian distribution. For any model g , we define its mean response vector by $\bar{g} := \frac{1}{n} \sum_{i=1}^n g(x_i) \in \mathbb{R}^{d_{\text{out}}}$, where d_{out} denotes the output dimensionality and $g(x_i)$ denotes either the model’s logits or its final-layer representations. We then define its disfavored dimensions as the set of bottom- m coordinates of $\bar{g}$:

$$\mathcal{M}_g := \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

We adopt the mean response vector as the bias estimator because it constitutes a more stable and noise-robust alternative to the per-sample argmin frequency (i.e., counting how often each coordinate attains the minimum value, as illustrated in Figure 2), while remaining empirically effective. A formal comparison of the two estimators is provided in Appendix C.1.

For any two models f and f' , we define their *identity dimensions* as the intersection $\mathcal{S} := \mathcal{M}_f \cap \mathcal{M}_{f'}$. If two models share an initialization lineage, they should independently recover similar sets of disfavored coordinates, yielding a non-trivial overlap of their bottom- m sets. The intersection-based criterion thus functions as a mutual-verification step: it attenuates spurious correlations attributable to architectural similarity or shared training dynamics and isolates bias signals more plausibly attributable to a shared initialization. When one model is closer to initialization, its disfavored coordinates may instead be used directly as an anchor set.

Correlation statistics. Restricting both models to the identity dimensions $\mathcal{S}$, we compare their responses across the random inputs.¹ For each $j \in \mathcal{S}$, we compute the Kendall–Tau rank correlation

$$\tau_j = \text{KendallTau}(\{f(x_i)_j\}_{i=1}^n, \{f'(x_i)_j\}_{i=1}^n).$$

This yields the set of correlation statistics $T = \{\tau_j\}_{j \in \mathcal{S}}$, which characterizes the distribution of agreement between the two models over the identity dimensions. Consistent with Section 3, same-lineage models exhibit distributionally higher agreement on these dimensions than unrelated models.

Hypothesis testing against a null distribution. Because the correlation distributions of same-lineage and different-lineage models can overlap, thresholding an individual statistic is insufficient. We instead test whether the observed correlations are significantly larger than would be expected under the null hypothesis of no lineage relation. Ideally, this null would be estimated from the correlation distribution between independent models, as in Figure 2. Estimating such a baseline empirically, however, requires comparisons across many independently initialized models and is therefore computationally expensive. As shown in Appendix B, the null correlation distribution between independently initialized models is well approximated by a Gaussian. This observation motivates a cheaper simulation-based alternative: an empirical null constructed by passing Gaussian surrogates through the identical pipeline, as summarized in Algorithm 1.

Although this simulation-based null already yields a strong empirical approximation, it incurs avoidable stochasticity and computational overhead: its precision is bounded by the number of simulation trials, its estimates vary across runs, and comparing two empirical distributions generally requires additional distributional assumptions imposed by the chosen test, e.g., the one-sided t -test or the Mann–Whitney U test. Because the null distribution of Kendall–Tau (under weak independence) admits a closed form, we further replace the simulation-based procedure with an analytical test, while providing both the empirical and the analytical implementations in our code repository. Formally, under the null hypothesis each Kendall–Tau correlation is approximately zero-mean with known variance σ^2 . By the central limit theorem, as the number of identity dimensions $|\mathcal{S}|$ grows, the distribution of $\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j$ is well approximated by $\bar{\tau} \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right)$. Although the softmax normalization induces weak dependencies across coordinates, these are attenuated by the relatively high temperature ($T = 10$), rendering the independence assumption an adequate approximation in practice. We empirically verify that this Gaussian approximation closely matches the distribution produced by the full pipeline (Appendix B). We accordingly compute the one-sided z -score $z = \frac{\bar{\tau}}{\sigma/\sqrt{|\mathcal{S}|}}$ and the associated p -value $p = 1 - \Phi(z)$, where Φ denotes the standard normal cumulative distribution function. We reject the null and declare shared lineage whenever $p < 0.01$.

Practical instantiation. The model output may be taken as either the logits or the hidden representations. Both choices are consistent with our formulation; however, using the logits requires substantially more random inputs to reliably estimate the bias patterns as the vocabulary size grows, which incurs considerable computational overhead (Appendix C.2 provides the supporting theoretical analysis). For efficiency and stability, we therefore adopt the final-layer hidden states as the default output. The resulting p -value constitutes a direct statistical test for lineage that does not rely on heuristic similarity thresholds.

¹Prior to computing the correlations, we apply a row-wise softmax normalization to each sample’s output vector. This eliminates scale discrepancies across models and coordinates while preserving the relative ordering that carries identity information for rank-based statistics such as Kendall–Tau. A relatively high temperature is employed to avoid degenerate one-hot behavior.

Algorithm 1 Distribution Correlation Test on Identity Dimensions

Require: base model f , suspicious model f' ; random inputs $X = \{x_i\}_{i=1}^n$; fingerprint size m ; significance level α

► **Step 1: Localize biased dimensions**

- 1: Compute average outputs $\bar{f}, \bar{f}'$ over X
- 2: $\mathcal{M}(f), \mathcal{M}(f') \leftarrow$ bottom- m dimensions of $\bar{f}, \bar{f}'$
- 3: $\mathcal{S} \leftarrow \mathcal{M}(f) \cap \mathcal{M}(f')$ (identity dimensions)

► **Step 2: Form correlation distribution**

- 4: Normalize $f(x_i)_{\mathcal{S}}, f'(x_i)_{\mathcal{S}}$ for each $x_i \in X$ across identity dimension
- 5: **for** each $s_j \in \mathcal{S}$ **do**
- 6: $\tau_j \leftarrow \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n)$ (correlation in input-wise preferences)
- 7: **end for**
- 8: $\mathcal{T} \leftarrow \{\tau_j\}_{j=1}^{|\mathcal{S}|}$

► **Step 3: Hypothesis test against null**

- 9: Construct $\mathcal{T}_{\text{null}}$ by applying the same pipeline to two Gaussian matrices $Y^{(1)}, Y^{(2)} \sim \mathcal{N}(0, I)^{N \times d_{\text{out}}}$
- 10: Test $H_0 : \mathcal{T} = \mathcal{T}_{\text{null}}$ vs. $H_1 : \mathcal{T} > \mathcal{T}_{\text{null}}$
- 11: **Return** SameLineage $\leftarrow \mathbf{1}(p\text{-value} < \alpha)$

5 EXPERIMENTS

The evaluation comprises two parts. In Section 5.1, we demonstrate that the proposed method functions as a genuine fingerprint by establishing that (i) it enables *birth verification at the seed level* and (ii) it remains *verifiable across the entire training lifecycle*. Existing baselines, by contrast, do not support verification at early pretraining stages and rely predominantly on data-dependent signals that collapse under substantial distribution shift. We report results obtained with the analytical null and *hidden-state* outputs; additional results obtained with the empirical null under both *logit* and *hidden-state* outputs, evaluated via the one-sided t -test and the Mann–Whitney U test, are deferred to Appendix D.2. In Section 5.3, we evaluate all methods under practical infringement scenarios using *LeaFBench* (Shao et al., 2025). Over 65 distinct model instances spanning 6 representative post-development techniques, our method matches the strongest baselines in post-training settings and remains robust to a diverse set of deployment transformations.

Baselines. We principally consider four passive fingerprinting baselines, each either weight-based or representation-based. *Intrinsic fingerprint* (Yoon et al., 2025) (denoted *PDF* in some works) compares two models by the similarity of the layerwise standard-deviation profiles of their attention parameters. *REEF* (Zhang et al., 2024) computes the centered-kernel-alignment (CKA) similarity between the feature representations elicited from identical inputs across the two models. *PCS* and *ICS* (Zeng et al., 2024) (collectively *HuRef* in some works) are weight-similarity methods: PCS flattens all parameters and measures their cosine similarity, whereas ICS constructs invariant terms from the weights and measures cosine similarity over those invariants. In Section 5.3 we additionally include a weaker gradient-based fingerprint, *Gradient* (Shao et al., 2025). Following Zhang et al. (2024), we adopt a **similarity threshold of 0.8** for binary decisions.

Throughout the experiment tables, cell color encodes ground-truth lineage: **green** denotes a same-source pair and **red** denotes a different-source pair. For instance, s_{42}^{init} vs. s_{42}^{base} compares a model initialized with seed 42 against its continued-pretrained descendant and is therefore green; conversely, s_{2000}^{init} vs. s_{42}^{base} compares a seed-2000 initialization against a model trained from a seed-42 initialization and is therefore red. The markers **✓** and **✗** denote a correct detection and a detection error, respectively.

5.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) equipped with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. The main paper reports results for the LLaMA-style models and defers those for the Qwen-style models to Appendix D; the overall conclusions are consistent across the two architectures.

Distinct initialization seeds produce distinct fingerprints. Table 1 reports the p -values of our correlation test for pairs of models initialized with different random seeds (42, 123, 1000, and 2000). Every p -value satisfies $p > 0.01$, so the test does not detect shared lineage between any cross-seed

Table 1: Fingerprint behavior for pairs of models initialized with distinct random seeds. Every p -value exceeds 0.01, so no cross-seed pair is flagged as sharing lineage.

Model Pairs	p -value (> 0.01)
s_{42} VS. s_{2000}	0.116
s_{123} VS. s_{42}	1.000
s_{1000} VS. s_{123}	0.327
s_{2000} VS. s_{1000}	1.000

Table 3: An identical corpus and data ordering do not render fingerprint behavior identical across distinct initializations; every cross-seed p -value exceeds 0.01.

Model Pair	p -value (> 0.01)
s_{42}^{init} VS. s_{123}^{base}	0.573
s_{123}^{init} VS. s_{2000}^{base}	0.724
s_{2000}^{init} VS. s_{1000}^{base}	0.109
s_{1000}^{init} VS. s_{42}^{base}	0.883

Table 2: A trained model exhibits the same fingerprint behavior as its initialization, with p -value < 0.01 for every seed-model pair.

Model Pairs	SeedPrints	Baselines			
	p -value	Intrinsic	REEF	PCS	ICS
s_{42}^{init} VS. s_{42}^{base}	3.37e-26 [✓]	-0.021 [×]	0.375 [×]	0.580 [×]	0.196 [×]
s_{123}^{init} VS. s_{123}^{base}	3.16e-33 [✓]	0.149 [×]	0.369 [×]	0.581 [×]	0.188 [×]
s_{1000}^{init} VS. s_{1000}^{base}	1.12e-21 [✓]	-0.252 [×]	0.381 [×]	0.581 [×]	0.188 [×]
s_{2000}^{init} VS. s_{2000}^{base}	3.70e-31 [✓]	-0.337 [×]	0.331 [×]	0.581 [×]	0.188 [×]

Table 4: Fingerprint persistence under continual training on heterogeneous corpora (base model: seed 1000, pretrained on openwebtext).

Setting	SeedPrints	Baselines			
Continual corpus (seed)	p -value	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	≈ 0 [✓]	1.000 [✓]	0.759 [×]	0.999 [✓]	0.996 [✓]
TinyStories (123)	1.000 [✓]	0.950 [×]	0.658 [✓]	0.332 [✓]	0.012 [✓]
the_stack (1000)	≈ 0 [✓]	0.489 [×]	0.557 [×]	0.585 [×]	0.123 [×]
the_stack (123)	1.000 [✓]	0.445 [✓]	0.580 [✓]	0.301 [✓]	0.026 [✓]

pair and thus reliably separates models initialized from different seeds. Distinct seeds therefore induce distinct fingerprint behaviors, permitting models to be separated “at birth.”

Training preserves the initialization fingerprint. Table 2 compares each initialization model s^{init} against its descendant s^{base} , obtained by training on the OpenWebText corpus (Gokaslan et al., 2019) (≈ 10 B tokens). For every seed-model pair the p -value satisfies $p < 0.01$, indicating that the bias profiles remain strongly correlated and thus share a common lineage; in other words, the trained model inherits the fingerprint of its initialization. The baseline methods, by contrast, uniformly fail to discriminate across seeds, which indicates that their separability originates in training-induced artifacts rather than in initialization.

Identical data and ordering do not cause fingerprints to converge. In Table 3, all four “suspect” models s_i^{base} with $i \in \{42, 123, 100, 2000\}$ are trained on *the identical corpus* (OpenWebText) under *the identical data ordering*; we fix the training seed to lock the ordering, so that the sole point of variation is the initialization seed i . Across every cross-seed pair the p -value remains > 0.01 , in sharp contrast to the near-zero values of Table 2. The fingerprint is therefore seed-specific even under identical data and curriculum.

Continual training on heterogeneous datasets does not confound the fingerprint. The preceding experiments serve solely to establish that SeedPrints can act as a biometric fingerprint. From a copyright standpoint, the inability of prior methods to distinguish models with different seeds is not itself a deficiency, since initialization seeds carry no well-defined copyright status. The consequential fragility is instead that their attribution is readily misled by the *data distribution, failing to recognize lineage when the training distribution shifts substantially during continued training*.

In Table 4, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two markedly different corpora: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We contrast (i) true descendants trained from the base against (ii) *distractors* derived from a different base model (seed 123). Because the baseline methods are insensitive to initialization differences, the distractor base is pretrained with a different data ordering on OpenWebText and then continued on the *same* target corpus to form the distractor. The question is whether an attribution method can identify which descendant genuinely shares lineage with the base.

Table 5: Fingerprinting results under large-scale fine-tuning. Each row compares a target model against LLaMA-2-7B. **SeedPrints** reports the p -value of our correlation test, where $p < 0.01$ indicates a strong lineage signal. The four baselines each report a similarity score (decision threshold 0.8; larger is better).

Model	# Tokens	SeedPrints p	Intrinsic $\uparrow$	REEF ($\uparrow$)	PCS ($\uparrow$)	ICS ($\uparrow$)
Llama-2-finance-7B (Heenan, 2023)	5M	10^{-41955} ✓	1.0000 ✓	0.9950 ✓	0.9979 ✓	0.9952 ✓
Vicuna-1.5-7B (Chiang et al., 2023)	370M	$10^{-103043}$ ✓	1.0000 ✓	0.9985 ✓	0.9985 ✓	0.9949 ✓
Wizardmath-7B (Luo et al., 2023)	1.8B	$10^{-180550}$ ✓	1.0000 ✓	0.9979 ✓	1.0000 ✓	0.9994 ✓
Meditron-7B (Chen et al., 2023)	48B	10^{-42566} ✓	0.9990 ✓	0.9978 ✓	1.0000 ✓	0.9817 ✓
CodeLlama-7B (Meta AI, 2024)	500B	10^{-3552} ✓	0.9480 ✓	0.9947 ✓	0.6863 ✗	0.3369 ✗
Llemma-7B (Azerbayev et al., 2023)	700B	10^{-5136} ✓	0.9470 ✗	0.9984 ✓	0.6682 ✗	0.2905 ✗

We observe that every prior baseline fails under the code setting (The Stack), misclassifying true descendants as distractors. This indicates that these methods primarily track domain similarity rather than lineage identity: TinyStories is distributionally closer to the pretraining corpus (OpenWebText), whereas The Stack diverges sharply, and a distribution shift of this magnitude readily evades detection. Our method, in contrast, correctly attributes lineage under both corpora. The proposed fingerprint is therefore not a proxy for the data distribution: it withstands substantial domain shift and persists beyond the initial pretraining stage.

5.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. Can the method reliably identify its offspring along the training trajectory? We first assess this in a controlled pretraining experiment on OpenWebText, in which the initialization seed is known, by testing lineage at intermediate checkpoints; we observe consistent detection at every stage (Appendix D.2.1, Figure 5).

Large-scale pretraining. To assess lineage under realistic foundation-model training, we further analyze ten checkpoints from the OLMo-2-7B (OLMo et al., 2024) Stage-1 run (5B $\rightarrow$ 3.9T tokens), treating the final checkpoint as the target model and each preceding checkpoint as a candidate ancestor. In contrast to the controlled OpenWebText setting, this run spans heterogeneous corpora, a substantially longer optimization horizon, and large-scale representation drift. We report $1 - p$ to preserve directional consistency with the similarity-based baselines; all p -values satisfy $p \ll 0.01$ and are numerically close to zero.

As shown in Figure 1, **all existing baselines fail to verify lineage during early pretraining** (within the first **trillion** tokens, $ICS \approx 0$, $PCS < 0.3$, and $REEF < 0.6$), whereas SeedPrints remains consistently verifiable and its signal strengthens monotonically throughout training. These results expose a critical gap in existing evaluation practice: **most prior work verifies lineage only after fine-tuning, the regime in which detection is substantially easier**. Our findings, however, indicate that large-scale pretraining can itself dramatically reshape representations and amplify lineage signals, thereby inducing a *false sense of safety*. We contend that lineage verification must prioritize the early stages of pretraining, at which misuse is hardest to detect, rather than relying exclusively on late-stage checkpoints.

Large-scale fine-tuning. We further compare our method against the baselines under standard fine-tuning-stage evaluations. Specifically, we test suspect models fine-tuned from the Llama-2-7b base model with data volumes spanning 5 million to 700 billion tokens. The suspects comprise diverse downstream variants, including Llama-2-finance-7b (Heenan, 2023), Vicuna-1.5-7b (Chiang et al., 2023), WizardMath-7b (Luo et al., 2023), Chinese-LLaMA-2-7b (Chen et al., 2023), CodeLlama-7b (Meta AI, 2024), and Llemma-7b (Azerbayev et al., 2023), with fine-tuning data volumes of 5M, 370M, 1.8B, 13B, 500B, and 700B tokens, respectively. As reported in Table 5, our method maintains $p < 0.01$ across all settings.

5.3 ROBUSTNESS UNDER REALISTIC DEPLOYMENTS

Real-world deployments routinely subject foundation models to *parameter-altering* adaptations (e.g., fine-tuning, PEFT, quantization) that can attenuate or distort fingerprint signals. Our method is designed to remain effective in this regime. To quantify its reliability under such conditions, we adopt *LeaFBench* (Shao et al., 2025), a benchmark for language-model copyright auditing.

Across source models. We consider two categories of source model for auditing: *Pre-Trained (PT)* and *Instruction-Tuned (IT)*. For each category, the task is to separate *derivative* models (produced by post-training adaptation of the source) from *independent* models (trained without access to the source weights). This mirrors the standard auditing scenario in which a lineage claim must be tested for either a PT base or its IT variant. The evaluation spans 65 models in total; the complete enumeration of model names and HuggingFace repositories appears in Appendix E.

Metrics. We quantify detection performance using (i) the Area Under the ROC Curve (AUC) and (ii) the Kolmogorov–Smirnov (KS) statistic (Berger & Zhou, 2014), the latter measuring the maximum separation between the score distributions of the derivative and independent models.

Our method yields a per-model p -value from a statistical test, a quantity fundamentally distinct from the similarity scores produced by the baselines. Whereas a similarity score is a linear measure on which differences carry proportional meaning, a p -value is a tail probability quantifying the rarity of an observation under the null hypothesis and thus cannot be interpreted on a linear scale. To enable comparison with baselines that compute AUC from similarity scores, we map each p -value to a score via $s = 1 - p$. This transformation is not fully faithful to the statistical semantics of the p -value and may be suboptimal for our method, since small p -values vary on an exponential scale while threshold sweeping proceeds linearly, which impedes fine-grained discrimination. Even so, our approach attains comparable or superior performance. Concretely, we (i) compute AUC for all methods by sweeping thresholds over the scores s and (ii) report the KS statistic as a threshold-free measure of distributional separability. This protocol maintains compatibility with the baseline pipelines while preserving the statistical interpretability of our method. We omit the Manhattan distance, as absolute distances between p -values are not meaningful.

We emphasize that, in most practical scenarios, calibrating an optimal similarity threshold is infeasible. Although we report AUC for comparability, our method differs from prior methods in kind: it constitutes a direct statistical test. **For any pair of models, our approach returns a p -value that supports a definitive same-lineage decision without recourse to a tunable threshold**, which ensures reliable verification in practice.

Parameter-altering techniques. To characterize fingerprinting robustness, we evaluate models produced by (1) instruction tuning (Instruct), (2) general-purpose fine-tuning (Finetune), (3) parameter-efficient fine-tuning (PEFT), (4) quantization, (5) model merging (Merge), and (6) distillation.

5.3.1 FAMILY-WISE EFFECTIVENESS

The pool contains seven model families. For each family, we treat both its pretrained base model (PT) and its instruction-tuned model (IT) as distinct source models and compare each against all remaining models in the pool of 65; the results are reported in Table 6. We exclude the TinyLlama-1.1B family from Table 6, as it constitutes a trivial case for all methods. Each column corresponds to a single family, for which AUC is computed over the associated tests using a family-calibrated threshold, whereas the overall score is computed across all test pairs under a single global threshold; consequently, the overall score does not equal the mean of the per-family scores. Three methods—*Intrinsic*, *ICS*, and our *SeedPrints*—are essentially saturated across all alteration types (overall AUC of 0.994, 0.994, and 0.992, respectively; overall KS of 0.989, 0.943, and 0.986), and their family-wise scores attain the maximum in most cases. The remaining two baselines, *REEF* and *Gradient*, perform substantially worse. These results indicate that, although our method is designed principally to capture intrinsic seed-level fingerprints, it operates as a biometric-like fingerprint that supports reliable and persistent identity tracking across diverse model families.

5.3.2 ALTERATION-WISE ROBUSTNESS

Across alteration types, *Instruct* is uniformly the easiest case, on which even the weakest baseline (*Gradient*) attains AUC 0.895, whereas *Finetune* and *Merge* are more challenging for all methods; under *Finetune*, in particular, no method attains perfect performance. The effects of *PEFT*, *Quantization*, and *Distillation* are more heterogeneous: *REEF* degrades appreciably under *Quantization* ($0.996 \rightarrow 0.871$) and *Distillation* ($0.996 \rightarrow 0.858$) yet remains resilient to *PEFT* (0.994), whereas *PEFT* and *Quantization* are the harder cases for *Gradient* ($0.895 \rightarrow 0.776/0.756$). The three stronger methods are largely robust to every transformation (all scores > 0.9). *SeedPrints* declines marginally to 0.959 on *Merge*, corresponding to a single misclassified merge model out of seventeen; although this amounts to only one error, it suggests that weight-space interpolation with unrelated model

Table 6: Performance of LLM fingerprinting methods across source models. “PT” and “IT” denote the use of the pre-trained model and the instruction-tuned model, respectively, as the source. The per-family breakdown covers the 6 families that provide both a PT and an IT model. The “Overall” scores are computed over all test model pairs and are not averages of the per-family scores.

Method	Metric	Qwen2.5-7B		Qwen2.5-14B		Llama3.1-8B		Mistral-7B		Gemma2-2B		Llama2-7B		Overall
		PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	
REEF	AUC $\uparrow$	0.796	0.862	0.947	0.913	1.000	0.999	0.997	0.997	0.963	0.968	0.750	0.706	0.915
	KS Statistic $\uparrow$	0.492	0.633	0.875	0.735	0.987	0.987	0.975	0.980	0.950	0.837	0.750	0.694	0.739
Gradient	AUC $\uparrow$	0.657	0.743	0.763	0.789	0.897	0.904	0.872	0.682	1.000	1.000	0.650	0.768	0.801
	KS Statistic $\uparrow$	0.354	0.423	0.500	0.563	0.646	0.705	0.725	0.500	1.000	1.000	0.275	0.479	0.508
ICS	AUC $\uparrow$	1.000	1.000	0.997	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	0.975	0.979	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.943
Intrinsic	AUC $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.989
SeedPrints	AUC $\uparrow$	0.992	0.994	0.988	0.990	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.992
	KS Statistic $\uparrow$	0.985	0.987	0.975	0.980	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.986

Table 7: Performance of LLM fingerprinting methods across parameter-altering transformation classes.

Method	Metric	Instruct	Finetune	PEFT	Quantization	Merge	Distillation	Overall Avg
REEF	AUC $\uparrow$	0.996	0.872	0.994	0.871	0.954	0.858	0.924
	KS Statistic $\uparrow$	0.949	0.637	0.940	0.812	0.851	0.615	0.801
Gradient	AUC $\uparrow$	0.895	0.729	0.776	0.756	0.832	0.887	0.812
	KS Statistic $\uparrow$	0.756	0.441	0.469	0.586	0.519	0.679	0.575
ICS	AUC $\uparrow$	1.000	0.987	1.000	1.000	0.981	1.000	0.995
	KS Statistic $\uparrow$	1.000	0.925	1.000	1.000	0.941	1.000	0.978
Intrinsic	AUC $\uparrow$	1.000	0.980	1.000	1.000	1.000	1.000	0.997
	KS Statistic $\uparrow$	1.000	0.971	1.000	1.000	1.000	1.000	0.995
SeedPrints	AUC $\uparrow$	1.000	0.995	0.990	1.000	0.959	1.000	0.991
	KS Statistic $\uparrow$	1.000	0.990	0.980	1.000	0.941	1.000	0.985

weights can dilute the initialization signal below the detection threshold. Averaged over all six alteration types, Intrinsic attains the highest overall AUC, followed closely by ICS and SeedPrints.

5.4 ABLATION STUDY ON HYPERPARAMETERS

The method has three hyperparameters: the number of random input sequences n , the length l of each random sequence, and the fingerprint size m . The first two jointly control the total volume of random inputs, whereas m specifies the number of low-mean coordinates retained for testing. Throughout the main experiments in Sections 5.2 and 5.3, we set $n=2000$, $l=1024$, and $m=0.1 d_{\text{out}}$, where d_{out} denotes the hidden-state dimensionality; expressing m relative to d_{out} renders the method directly transferable across open-source models of differing width. Here we vary each hyperparameter in turn while holding the remaining two at their default values ($n = 2000$, $m = 400$, $l = 768$), and report the empirical accuracy and false-positive rate at a fixed significance level $\alpha = 0.01$ in Table 8.

Increasing n or l consistently improves identification performance. This behavior is predicted by the theoretical analysis in Section C.2, which establishes that accurate recovery of the population-level bias requires the number of queries to satisfy $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$, where γ is the bias margin separating the biased coordinates from the remaining dimensions. Thus each target output coordinate demands a minimum sample budget for reliable detection, which explains the suboptimal performance at small n . Because γ increases with l through accumulation effects, longer sequences enlarge the separation; consequently, at a fixed (insufficient) number of sequences, increasing l lowers the required query count and improves detection.

Table 8: Ablation study under significance level $\alpha = 0.01$.

Ablation on n	200	500	2000
ACC	0.7368	0.7778	0.9375
Empirical FPR	0	0	0
Ablation on l	512	768	1024
ACC	0.9048	0.9375	1.0000
Empirical FPR	0.04	0	0
Ablation on m	200	400	800
ACC	0.9375	0.9375	0.875
Empirical FPR	0.04	0	0.1463

The dependence on m is non-monotonic. When m is too small, the selected coordinates are unstable, as there are too few empirical observations to form a reliable distribution. When m is too large, the inclusion of many irrelevant coordinates injects noise that shrinks γ and, by the sample-complexity bound, would require quadratically more samples to compensate. Empirically, m in the range 200–400 performs well across model sizes, in agreement with the heuristic $m \approx 0.1 d_{\text{out}}$ for typical $d_{\text{out}} \in [3000, 5000]$.

6 CONCLUSION

In this work we introduced *SeedPrints*, a stronger and intrinsic notion of LLM fingerprinting that traces a model’s lineage back to its random initialization. We established that initialization alone deposits a persistent “Galtonian” fingerprint on neural language models: untrained models exhibit stable, seed-dependent token-preference patterns, and these biases persist—both statistically and measurably—throughout training. On this basis, we formulated a distribution-correlation test over *identity dimensions* that detects shared lineage at a calibrated significance level. Across the LLaMA- and Qwen-style families and a broad suite of realistic deployments (continued pretraining on divergent corpora, instruction tuning, PEFT, quantization, merging, and distillation), SeedPrints enables “birth-to-lifecycle” verification and remains robust under domain shift, thereby providing a practical instrument for provenance and copyright auditing.

Limitations and Future Directions This work targets fingerprint signals rooted in initialization-stage bias. The proposed method requires access to model outputs (logits or hidden states) and is therefore inapplicable when only restricted API access is available. We nonetheless emphasize that the pursuit of *true fingerprints* for LLMs remains at an early stage: most existing approaches, and black-box methods in particular, rely on signals that emerge only in later training stages or from alignment-driven artifacts, and thus cannot robustly protect a model at the pretrained stage. Reliable all-stage fingerprints are accordingly all the more urgently needed. Full discussions are provided in Section A.2.

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788
789
790
791
792
793
794
795
796
797
798
799
800
801
802
803
804
805
806
807
808
809

Appendix Contents

A	LLM Usage and Limitation Discussions	16
A.1	LLM Usage	16
A.2	Limitations and Future Directions	16
B	Validation of the Empirical and Analytical Null Distribution	16
B.1	Empirical True Null from Independently Initialized Models	16
B.2	Gaussian Surrogate as a Simulation-Based Approximation	16
B.3	Analytical Approximation under Weak Dependence	17
C	Theoretical Properties of Our Methods	18
C.1	Stability Analysis of Mean-Based Identity Extraction vs. Per-Sample Argmin Voting	19
C.1.1	Stability of Sample-Mean-Based Extraction	19
C.1.2	Instability of Per-Sample Argmin Voting	20
C.2	Influence of Hyperparameters	22
D	More Experiment Details	23
D.1	Implementation Details and Licenses	23
D.2	Complementary SeedPrints Results using simulation-based null	23
D.2.1	Birth-to-Lifecycle “Biometric” Fingerprinting	23
D.2.2	All-stage verifiable fingerprints	25
D.3	Cross-Size Fingerprint Within the Qwen-2.5 series	26
E	Model Catalog and Decoding Settings	26
E.1	Model list by family	27
E.1.1	Qwen-2.5-7B	27
E.1.2	Qwen2.5-14B	27
E.1.3	Llama-3.1-8B	27
E.1.4	Mistral-7B-v0.3	28
E.1.5	Gemma-2-2B	28
E.1.6	Llama-2-7B	28
E.1.7	TinyLlama-1.1B	28
E.2	Sample Results	29

A LLM USAGE AND LIMITATION DISCUSSIONS

A.1 LLM USAGE

We employ AI assistants, specifically ChatGPT and Gemini, for writing and formatting support. Their use is confined to grammar and style checking, improving the clarity of figure and table captions, and other surface-level edits. For programming-related tasks, we occasionally use GitHub Copilot and Claude as coding assistants, e.g., for code auto-completion and debugging suggestions.

A.2 LIMITATIONS AND FUTURE DIRECTIONS

Although this work identifies seed-born, training-persistent biases that serve as fingerprint signals, the proposed method operates within the white-box paradigm. It is inapplicable when only restricted API access is available, for instance when responses are deterministic or safety-filtered. We nonetheless emphasize that the pursuit of *true fingerprints* for LLMs remains at a very early stage, even under the white-box setting. Most existing fingerprinting approaches depend on training-induced signatures or alignment behaviors and therefore cannot robustly protect pretrained or early-stage models. By analogy to human fingerprints, which remain stable irrespective of age or environment, we contend that a true LLM fingerprint should persist across continued training and deployment contexts rather than encode incidental training artifacts. Attaining this objective—particularly under black-box access, where a substantial performance gap relative to white-box methods persists—remains a challenging open direction that demands substantial further work. We hope this study motivates continued research toward principled and robust fingerprinting frameworks for large models.

B VALIDATION OF THE EMPIRICAL AND ANALYTICAL NULL DISTRIBUTION

The hypothesis test of Section 4 decides whether two models share lineage by comparing the distribution of their correlation statistics over random inputs against a null distribution. This null distribution models the correlation statistics between two *independent models evaluated on random inputs*.

We examine two approximations of this null distribution: (i) an empirical, simulation-based null obtained by applying the full pipeline to Gaussian surrogate outputs, and (ii) an analytical null derived from the known distribution of Kendall–Tau correlations. In this section we justify both approximations in three steps: characterizing the true null via independently initialized models, introducing the Gaussian surrogate as a tractable simulation-based proxy, and validating the analytical approximation.

B.1 EMPIRICAL TRUE NULL FROM INDEPENDENTLY INITIALIZED MODELS

The ideal null distribution is the distribution of the correlation statistics between two independent models evaluated on random inputs. The most direct instantiation of the requisite independence is a comparison of two independently initialized models (with different random seeds and no training). We estimate this empirical null by evaluating ~ 2500 such pairs and computing Kendall–Tau correlations over 10,000 random inputs, as in Section 3. Figure 3 shows that the resulting true null distribution is empirically Gaussian and centered at zero.

B.2 GAUSSIAN SURROGATE AS A SIMULATION-BASED APPROXIMATION

Directly estimating the null distribution from independently initialized models is computationally expensive, as it requires access to many models and repeated evaluations. To obtain a tractable alternative, we construct a simulation-based null using Gaussian surrogates as proxies for model outputs under the null.

This choice rests on two considerations. First, approximating neural-network outputs by Gaussian distributions is a well-established practice in the auditing literature, supported by both empirical and theoretical studies (Carlini et al., 2022; Lee et al., 2019). Second, our pipeline depends solely on relative-ordering and selection operations (rank correlations and bottom- m selection) and does not exploit any semantic structure in the inputs; moreover, randomly initialized models possess

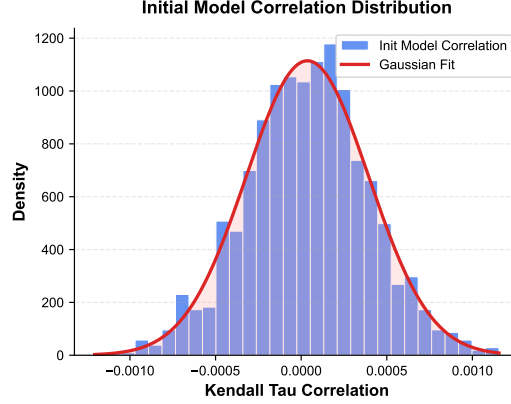


Figure 3: The empirical correlation distribution over ~ 2500 pairs of independently initialized models, evaluated on random inputs, closely matches a Gaussian distribution centered at zero.

independent parameter matrices, which implies that their outputs under the null exhibit no strong structured dependencies. Sampling independent Gaussian matrices therefore constitutes a simple and tractable proxy for the null behavior of independently initialized models.

We construct the surrogate null by sampling Gaussian matrices and applying the full pipeline, comprising identity-dimension selection and correlation computation:

1. Sample two Gaussian matrices of shape $\mathbb{R}^{N \times d_{\text{out}}}$,
2. Extract the most biased dimensions by selecting the bottom- m ranked average dimensions,
3. Take the intersection set $\mathcal{S}$ and compute correlations on it.

By applying the identical selection procedure to Gaussian samples, this simulation-based construction faithfully reproduces the dependency patterns induced by the pipeline itself (e.g., those arising from selecting $\mathcal{S}$ across dimensions). As shown in Figure 4 (middle), the resulting distribution matches the empirical null obtained from independently initialized models in its centered Gaussian shape, confirming that the Gaussian surrogate is an effective simulation-based approximation of the true null.

B.3 ANALYTICAL APPROXIMATION UNDER WEAK DEPENDENCE

Although the Gaussian surrogate captures the behavior of the null distribution under the full pipeline, it still requires repeated sampling and introduces estimation randomness. We therefore additionally derive the most efficient, closed-form analytical approximation.

Under the null hypothesis that the two models are independent², each Kendall–Tau correlation τ_j is approximately zero-mean with known variance σ^2 . If correlations across identity dimensions were independent, the central limit theorem would imply

$$\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right).$$

In practice, the softmax normalization introduces dependencies across dimensions. Nevertheless, Figure 4 demonstrates that the analytical Gaussian closely matches the distribution produced by the Gaussian surrogate pipeline across a range of settings, including varying numbers of random inputs n and output dimensions D . This indicates that the induced dependencies are sufficiently weak to leave the accuracy of the approximation materially unaffected. A plausible mechanism is that the relatively high temperature ($T = 10$) smooths the output distribution and thereby attenuates coupling between dimensions.

²As discussed in Section B.2, although outputs may exhibit weak dependencies due to shared inputs, this approximation remains accurate empirically.

Collectively, these results justify the analytical Gaussian null of Section 4 as an efficient, deterministic, and accurate approximation that obviates the need for costly empirical baseline construction.

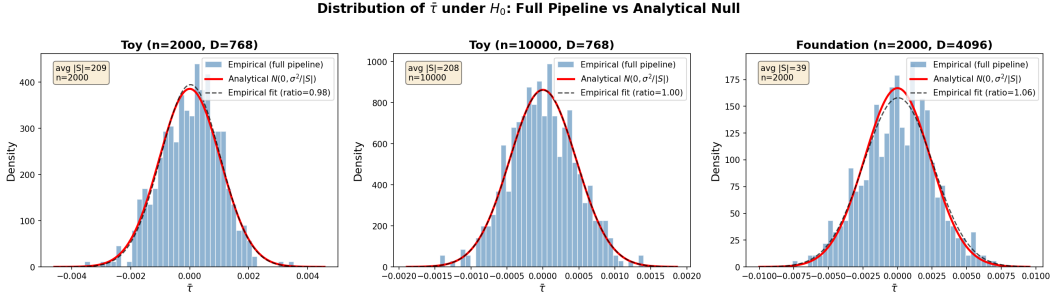


Figure 4: Validation of the analytical Gaussian null. We apply the full pipeline (identity-dimension selection and correlation aggregation) to Gaussian surrogate outputs and compare the resulting distribution of $\bar{\tau}$ against the analytical Gaussian $\mathcal{N}(0, \sigma^2/|S|)$. The close agreement indicates that the analytical approximation accurately captures the null behavior despite the weak dependencies introduced by the pipeline.

On variance scaling. The true null obtained from randomly initialized models has slightly smaller variance than the Gaussian null approximation. This discrepancy does not affect our hypothesis-test results. Our primary evaluation for the simulation-based null in Section D.2 employs the Mann–Whitney U -test, which is rank-based and hence invariant to any positive rescaling: for any $c > 0$, $x_i < x_j \Leftrightarrow cx_i < cx_j$, so all rank statistics and rejection thresholds are preserved. For tests that do depend on the absolute variance, we additionally evaluate variance-sensitive statistics such as the t -test, whose statistic transforms as $t' = t/c$ under a variance reduction by factor c . Since $c < 1$ in our setting, this rescaling shrinks the magnitude of t and renders the test more conservative rather than inflating significance. Empirically, across the extensive experiments of Section D.2, we find that the variance scaling does not affect the power of the t -test, which indicates that the distributional separation between the alternative and the null substantially exceeds the scaled variance.

For the analytical null used in the main paper, the variance enters explicitly in the normalization of the z -score, so any scale discrepancy is absorbed into the denominator. Because the analytical variance is slightly larger than the empirical one, the resulting z -scores are smaller, which makes the test conservative and suppresses false positives.

C THEORETICAL PROPERTIES OF OUR METHODS

Our objective is to identify the output coordinates that the initialized model is least likely to predict under random inputs. These coordinates correspond to directions the model intrinsically disfavors as a consequence of initialization-induced bias, independently of the input distribution, and form the basis of a stable fingerprint signal.

Setup. Consider a model g with fixed parameters and random inputs x_i (uniform random tokens or random embeddings, as in the main paper). For each output coordinate $j \in [d_{\text{out}}]$, $g_j(x)$ is therefore a random variable with population mean $\bar{g}_j := \mathbb{E}[g_j(x)]$. The population-level directions we seek to capture are the m output coordinates with the smallest expected values, defined formally as

$$\mathcal{M} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

Let $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ denote the sorted means in non-decreasing order, and write

$$\Gamma(m) := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$$

for the population “gap” between the m -th and $(m+1)$ -th smallest means. This gap quantifies the separation of the target set $\mathcal{M}$ from the remaining coordinates. Crucially, for a fixed m and input distribution, $\Gamma(m)$ is a function solely of the model’s intrinsic initialization bias. *Increasing the*

length of the random input sequences amplifies this bias via concentration, so that the separation becomes more pronounced at the population level.

Two natural empirical strategies exist for extracting these population-level least-likely coordinates.

1. **Sample-mean-based extraction (ours).** Estimate the expectation of each output coordinate by its empirical mean over the random inputs $X = \{x_i\}_{i=1}^n$ and select the m coordinates of smallest value. Following the notation of Section 4, in which g denotes the model (either the base model f or the suspect model f'):

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i), \quad \widehat{\mathcal{M}}^{\text{mean}} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \hat{g}_j.$$

2. **Per-sample argmin voting.** For each random input $x_i \in \mathbb{R}^{\ell \times d}$, select the smallest-value coordinate, then retain the m most frequently selected coordinates:

$$j^*(x_i) = \arg \min_{j \in \{1, \dots, d_{\text{out}}\}} g_j(x_i), \quad \widehat{\mathcal{M}}^{\text{vote}} = \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{i=1}^n \mathbf{1}\{j^*(x_i) \in J\}.$$

Let $p_j := \Pr(j^*(x) = j)$ denote the population probability that index j attains the minimum, and let

$$\mathcal{M}_{\text{vote}} := \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} p_j$$

be the population top- m set under voting.

Both strategies are intuitively reasonable. As we establish below, however, the sample-mean-based strategy is provably stable, whereas per-sample argmin voting is a discontinuous operator whose output can flip under arbitrarily small perturbations, rendering it inherently noise-sensitive.

C.1 STABILITY ANALYSIS OF MEAN-BASED IDENTITY EXTRACTION VS. PER-SAMPLE ARGMIN VOTING

C.1.1 STABILITY OF SAMPLE-MEAN-BASED EXTRACTION

We first show that the sample-mean-based operator, which selects the m smallest coordinates of the empirical mean vector, is stable under small perturbations to the model outputs. Such perturbations model realistic variations in the outputs of a *single* model, e.g., numerical noise or evaluation under differing precision modes.

Formally, we write the perturbed output as $g'(x) = g(x) + \eta(x)$, where $\eta(x) \in \mathbb{R}^{d_{\text{out}}}$ denotes the stochastic deviation. We assume that, for each sample x_i and coordinate j , the random variables $\eta_j(x_i)$ are independent across samples (though not necessarily across coordinates) and σ -sub-Gaussian with $\mathbb{E}[\eta_j(x_i)] = 0$.³

Lemma C.1 (Stability of sample-mean-based extraction). *Let $\hat{g}$ and $\hat{g}'$ be the empirical means over $X = \{x_i\}_{i=1}^n$, i.e., $\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i)$ and $\hat{g}' = \frac{1}{n} \sum_{i=1}^n g'(x_i) = \hat{g} + \frac{1}{n} \sum_{i=1}^n \eta(x_i)$. Let $\widehat{\Gamma}(m)$ denote the empirical gap at rank m . Then,*

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

Interpretation. The lemma furnishes a quantitative upper bound on the probability that the sample-mean-based extractor alters its selected set under perturbation. This instability probability decays exponentially in the number of averaged samples n , a consequence of the variance reduction from σ^2 per sample to σ^2/n in the empirical mean.

³This sub-Gaussian assumption is mild: bounded random variables are σ -sub-Gaussian (Hoeffding's lemma), and common initialization schemes together with bounded activations yield light-tailed behavior in both initialized and trained networks. The Gaussian case $\eta_j(x_i) \sim \mathcal{N}(0, \sigma^2)$ is a special instance of this assumption.

Proof. We condition on the sample $X = \{x_i\}_{i=1}^n$ throughout the proof, so that $\hat{g}$ and $\hat{\Gamma}(m)$ are fixed, and all probabilities are taken over the randomness of $\{\eta(x_i)\}_{i=1}^n$ only.

For any coordinate j , by definition of $\hat{g}'$, $\hat{g}'_j - \hat{g}_j = \frac{1}{n} \sum_{i=1}^n \eta_j(x_i)$. Each $\eta_j(x_i)$ is mean-zero and σ -sub-Gaussian, so their average is mean-zero and $(\sigma/\sqrt{n})$ -sub-Gaussian: the variance proxy scales as $1/n$ under averaging. Hence, for any $\varepsilon > 0$, by the standard sub-Gaussian tail bound,

$$\Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (1)$$

Now consider the event that *any* coordinate fluctuates by more than ε :

$$\left\{ \max_{1 \leq j \leq d_{\text{out}}} |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \right\}.$$

Taking a union bound over all d_{out} coordinates yields

$$\Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (2)$$

We now show that

$$\left\{ \widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g) \right\} \subseteq \left\{ \max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \right\}.$$

Equivalently, if $\max_j |\hat{g}'_j - \hat{g}_j| < \hat{\Gamma}(m)/2$, then the bottom- m identity set remains unchanged.

Let $j_{(1)}, \dots, j_{(d_{\text{out}})}$ be a permutation of $\{1, \dots, d_{\text{out}}\}$ such that $\hat{g}_{j_{(1)}} \leq \hat{g}_{j_{(2)}} \leq \dots \leq \hat{g}_{j_{(d_{\text{out}})}}$. Recall that the empirical gap at rank m is $\hat{\Gamma}(m) := \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} > 0$. Denote $\widehat{\mathcal{M}}^{\text{mean}}(g) := \{j_{(1)}, \dots, j_{(m)}\}$ and $\widehat{\mathcal{M}}_c^{\text{mean}}(g) := \{j_{(m+1)}, \dots, j_{(d_{\text{out}})}\}$. For any $i \in \widehat{\mathcal{M}}^{\text{mean}}(g)$ and any $\ell \in \widehat{\mathcal{M}}_c^{\text{mean}}(g)$, $\hat{g}_\ell - \hat{g}_i \geq \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} = \hat{\Gamma}(m)$. With $|\hat{g}'_i - \hat{g}_i| < \hat{\Gamma}(m)/2$ and $|\hat{g}'_\ell - \hat{g}_\ell| < \hat{\Gamma}(m)/2$, we have

$$\hat{g}'_\ell - \hat{g}'_i \geq (\hat{g}_\ell - \frac{\hat{\Gamma}(m)}{2}) - (\hat{g}_i + \frac{\hat{\Gamma}(m)}{2}) \geq \hat{\Gamma}(m) - \hat{\Gamma}(m) = 0.$$

Thus no coordinate in $\widehat{\mathcal{M}}_c^{\text{mean}}(g)$ can become smaller than any coordinate in $\widehat{\mathcal{M}}^{\text{mean}}(g)$ under this deviation bound, and therefore

$$\widehat{\mathcal{M}}^{\text{mean}}(g') = \widehat{\mathcal{M}}^{\text{mean}}(g).$$

Since $\hat{\Gamma}(m)$ is fixed under conditioning on X , plugging $\varepsilon = \hat{\Gamma}(m)/2$ into (2) yields

$$\Pr\left(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)\right) \leq \Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \mid X\right) \leq 2d_{\text{out}} \exp\left(-\frac{n\hat{\Gamma}(m)^2}{8\sigma^2}\right).$$

□

C.1.2 INSTABILITY OF PER-SAMPLE ARGMIN VOTING

In contrast, per-sample voting applies a discontinuous winner-take-all operator to each input, so arbitrarily small output perturbations can flip its decisions with non-vanishing probability. We again adopt the stochastic perturbation model $g'(x) = g(x) + \eta(x)$.

Lemma C.2 (Instability of per-sample argmin voting). *For a fixed input x , let*

$$j^*(x; g) := \arg \min_j g_j(x), \quad \Delta_g(x) := \min_{j \neq j^*(x; g)} \{g_j(x) - g_{j^*(x; g)}(x)\} > 0$$

denote the argmin and its margin. Then, for any fixed x ,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right),$$

and for n i.i.d. inputs $x_1, \dots, x_n$,

$$\Pr\left(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)\right) \geq 1 - (1 - \alpha)^n,$$

where $\alpha := \mathbb{E}_x \left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$.

Interpretation. Unlike mean extraction, voting derives no benefit from variance reduction: each input triggers a hard decision on the noisy outputs, so increasing n merely multiplies the opportunities for an index flip.

Proof. We first bound the per-input change probability $\Pr(j^*(x; g') \neq j^*(x; g) \mid x)$. Fix x and write $j^* := j^*(x; g)$ for brevity. By definition of the margin, $\Delta_g(x) = \min_{j \neq j^*} \{g_j(x) - g_{j^*}(x)\}$. Let $j^{(2)}$ be an index attaining this minimum, i.e., $g_{j^{(2)}}(x) - g_{j^*}(x) = \Delta_g(x)$.

Under the perturbation $g'(x) = g(x) + \eta(x)$, we have

$$g'_{j^{(2)}}(x) - g'_{j^*}(x) = [g_{j^{(2)}}(x) - g_{j^*}(x)] + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] = \Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)].$$

Since $\eta_{j^{(2)}}(x)$ and $\eta_{j^*}(x)$ are independent $\mathcal{N}(0, \sigma^2)$ variables, their difference is Gaussian with

$$\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \sim \mathcal{N}(0, 2\sigma^2).$$

The event that the argmin changes, $j^*(x; g') \neq j^*(x; g)$, certainly occurs whenever $g'_{j^{(2)}}(x) \leq g'_{j^*}(x)$. Hence,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x).$$

Using the expression above,

$$\Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x) = \Pr(g'_{j^{(2)}}(x) - g'_{j^*}(x) \leq 0 \mid x) = \Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x).$$

Let $Z \sim \mathcal{N}(0, 1)$. Since $\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \stackrel{d}{=} \sqrt{2}\sigma Z$, we obtain

$$\Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x) = \Pr\left(Z \leq -\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) = \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Therefore, $\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)$.

Now consider n i.i.d. inputs $x_1, \dots, x_n$. Conditional on the inputs, the events

$$E_i := \{j^*(x_i; g') \neq j^*(x_i; g)\} \quad E_i^c := \{j^*(x_i; g') = j^*(x_i; g)\}$$

are independent across i , because the perturbations $\eta(x_i)$ are independent across i . Moreover,

$$\Pr(E_i \mid x_i) = \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right).$$

Define $q(x_i) := \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x)$, and $\alpha := \mathbb{E}_x\left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)\right]$. Taking expectations over the randomness of x_i , we obtain $\mathbb{E}[\Pr(E_i \mid x_i)] = \alpha$. Therefore, by independence across i ,

$$\Pr(E_1^c \cap \dots \cap E_n^c) = \mathbb{E}\left[\prod_{i=1}^n (1 - q(x_i))\right] \leq \mathbb{E}\left[\prod_{i=1}^n (1 - \Phi(-\Delta_g(x_i)/(\sqrt{2}\sigma)))\right] \leq (1 - \alpha)^n,$$

where the last inequality uses i.i.d. inputs and Jensen's inequality applied to the convex function $u \mapsto \log(1 - u)$ on $u \in [0, 1)$. Therefore,

$$\Pr(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)) = 1 - \Pr(E_1^c \cap \dots \cap E_n^c) \geq 1 - (1 - \alpha)^n.$$

Finally, note that the empirical voting set $\widehat{\mathcal{M}}^{\text{vote}}$ is determined by the empirical counts of the dimensions $\{j^*(x_i; g)\}_{i=1}^n$. If the empirical gap between the m -th and $(m+1)$ -th most frequent dimensions is of order $O(1/n)$ (i.e., only a constant number of votes), then a change in $j^*(x_i)$ for a single input x_i is sufficient to swap the ordering across this boundary and thus alter the top- m set. Combining this observation with the lower bound $1 - (1 - \alpha)^n$ shows that, in such regimes, the probability that $\widehat{\mathcal{M}}^{\text{vote}}(g') \neq \widehat{\mathcal{M}}^{\text{vote}}(g)$ is bounded away from zero uniformly in n , i.e., per-sample argmin voting does not enjoy variance reduction with more samples. $\square$

C.2 INFLUENCE OF HYPERPARAMETERS

We now quantify the number of random inputs required for the empirical mean estimator to recover the true bias. We reuse the notation of the **Setup**: each output coordinate $g_j(x)$ is a random variable under random inputs $x \sim \mathcal{D}$, with population mean $\bar{g}_j := \mathbb{E}[g_j(x)]$; $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ denotes the means sorted in non-decreasing order, and $\gamma := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$ is the separation margin. We estimate $\bar{g}$ from n i.i.d. random inputs:

$$\hat{g}_j := \frac{1}{n} \sum_{i=1}^n g_j(x_i), \quad \hat{g} = (\hat{g}_1, \dots, \hat{g}_{d_{\text{out}}}).$$

The target dimensions set is $\mathcal{M}$ with size m and our empirical estimation gives $\widehat{\mathcal{M}}$.

Lemma C.3 (Sufficient condition for recovering the true set). *If $\|\hat{g} - \bar{g}\|_\infty \leq \varepsilon$ and $2\varepsilon < \gamma$, then the empirical bottom- m set equals the true set: $\widehat{\mathcal{M}} = \mathcal{M}$.*

Proof. For any $j \in \mathcal{M}$ and $j' \notin \mathcal{M}$, we have $\bar{g}_j \leq \bar{g}_{(m)}$ and $\bar{g}_{j'} \geq \bar{g}_{(m+1)}$. The ℓ_∞ bound gives

$$\hat{g}_j \leq \bar{g}_{(m)} + \varepsilon, \quad \hat{g}_{j'} \geq \bar{g}_{(m+1)} - \varepsilon.$$

Thus,

$$\hat{g}_j - \hat{g}_{j'} \leq -(\gamma - 2\varepsilon) < 0,$$

so no outside index can enter nor inside index leave the bottom- m set. $\square$

We next translate this recovery condition into a statement about the hyperparameters. We again invoke the sub-Gaussian assumption of Section C.1, which is mild: by Hoeffding's inequality any bounded random variable is σ -sub-Gaussian with σ proportional to its range. In our setting each $g_j(x)$ is a logit or a final hidden activation, quantities that are typically bounded or light-tailed in both initialized networks (owing to controlled parameter initialization) and trained networks.

Corollary C.4 (Sample complexity under sub-Gaussian coordinates). *Suppose each $g_j(x)$ is σ -sub-Gaussian. Then with probability at least $1 - \delta$,*

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Consequently, if

$$n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right),$$

then $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$.

Proof. For fixed j , sub-Gaussian concentration implies

$$\Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2 \exp(-n\varepsilon^2/2\sigma^2).$$

A union bound over all d_{out} coordinates yields

$$\Pr\left(\|\hat{g} - \bar{g}\|_\infty > \varepsilon\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

Set the right-hand side to δ and solve for ε to obtain, with probability at least $1 - \delta$,

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Now choose

$$\varepsilon := \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

If $n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right)$ (i.e., if $2\varepsilon < \gamma$), then Lemma C.3 applies and implies $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$. $\square$

Table 9: Fingerprint behavior under distinct random initializations, reported separately by model family.

(a) LLaMA-style models					(b) Qwen-style models				
Seed Pair	Logits Output		Hidden State		Seed Pair	Logits Output		Hidden State	
	<i>t-test</i>	<i>U-test</i>	<i>t-test</i>	<i>U-test</i>		<i>t-test</i>	<i>U-test</i>	<i>t-test</i>	<i>U-test</i>
s_{42} vs. s_{2000}	0.404	0.456	0.357	0.532	s_{123} vs. s_{1000}	0.741	0.727	0.094	0.074
s_{123} vs. s_{42}	0.214	0.295	0.678	0.565	s_{1000} vs. s_{123}	0.954	0.971	0.125	0.094
s_{1000} vs. s_{123}	0.219	0.246	0.363	0.335	s_{42} vs. s_{2000}	0.273	0.360	0.451	0.529
s_{2000} vs. s_{1000}	0.282	0.291	0.434	0.481	s_{2000} vs. s_{42}	0.215	0.206	0.230	0.295

Table 10: (LLaMA-style models) A trained model exhibits the same fingerprint behavior as its initialization, with p -value < 0.01 for every pair.

Model Pair	Logits Output		Hidden State		Baselines			
	<i>t-test</i>	<i>u-test</i>	<i>t-test</i>	<i>u-test</i>	Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{42}^{base}	3.33e-3✓	1.02e-3✓	2.20e-8✓	6.28e-8✓	-0.021×	0.375×	0.580×	0.196×
s_{123}^{init} vs. s_{123}^{base}	2.06e-3✓	7.33e-3✓	7.09e-6✓	1.37e-5✓	0.149×	0.369×	0.581×	0.188×
s_{1000}^{init} vs. s_{1000}^{base}	2.44e-3✓	4.14e-3✓	5.58e-4✓	2.81e-3✓	-0.252×	0.381×	0.581×	0.188×
s_{2000}^{init} vs. s_{2000}^{base}	5.63e-3✓	6.76e-3✓	4.00e-10✓	1.27e-9✓	-0.337×	0.331×	0.581×	0.188×

Accurate recovery therefore requires $n = \Omega\left(\frac{\log d_{out}}{\gamma^2}\right)$: the number of random queries grows logarithmically in the output dimension and inversely in the squared separation margin γ . The margin γ is itself governed by two factors:

- **Intrinsic bias strength.** Longer random input sequences amplify the initialization-induced bias via concentration, enlarging the population separation between coordinates.
- **Choice of m .** Distinct values of m select distinct regions of the ranked spectrum $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{out})})$ and thus induce distinct effective margins in practice.

The former is controlled through the input length l , whereas the latter is selected empirically from the observed gaps in the mean spectrum.

D MORE EXPERIMENT DETAILS

D.1 IMPLEMENTATION DETAILS AND LICENSES

We train all models with the Hugging Face Transformers Trainer (Wolf et al., 2020) and use Accelerate (Gugger et al., 2022) for distributed execution. Every open-source model is loaded from its official Hugging Face release and used under its original license: the Llama models under the Meta Llama Community License and the remaining models under Apache-2.0. All datasets are obtained via the Hugging Face Datasets library (itself Apache-2.0-licensed), and dataset content is governed by each dataset’s stated license.

D.2 COMPLEMENTARY SEEDPRINTS RESULTS USING SIMULATION-BASED NULL

This section reports the complementary results for Section 5. We evaluate the empirical simulation-based null of Algorithm 1 under both *logit* and *hidden-state* outputs, and we test using both the one-sided *t*-test (*t*-test) and the Mann–Whitney *U* test (*U*-test).

D.2.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) equipped with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. Because the simulation-based random baseline is stochastic, we report p -values averaged over 10 independent trials at a significance level

Table 11: (Qwen-style models) A trained model exhibits the same fingerprint behavior as its initialization, with p -value < 0.01 for every pair.

Model Pair	Logits Output		Hidden State	
	t -test	U -test	t -test	U -test
s_{123}^{init} vs. s_{123}^{base}	2.38e-03	1.68e-03	7.36e-15	3.38e-13
s_{1000}^{init} vs. s_{1000}^{base}	1.35e-04	4.84e-05	4.41e-13	2.01e-11
s_{42}^{init} vs. s_{42}^{base}	1.39e-03	1.46e-03	1.06e-24	2.05e-19
s_{2000}^{init} vs. s_{2000}^{base}	1.80e-03	1.28e-03	4.87e-24	1.92e-20

Table 12: Fingerprint persistence under continual training on heterogeneous corpora (base model: seed 1000, pretrained on openwebtext). “ U -test” denotes the Mann–Whitney U test.

(a) LLaMA-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	0✓	0✓	0✓	7.77e-89✓	1.000✓	0.759×	0.999✓	0.996✓
TinyStories (123)	1.000✓	1.00✓	0.943✓	0.902✓	0.950×	0.658✓	0.332✓	0.012✓
the_stack (1000)	0✓	1.73e-287✓	0✓	3.09e-69✓	0.489×	0.557×	0.585×	0.123×
the_stack (123)	0.616✓	0.479✓	0.732✓	0.831✓	0.445✓	0.580✓	0.301✓	0.026✓

(b) Qwen-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	5.36e-09✓	1.92e-07✓	8.49e-214✓	5.09e-71✓	1.000✓	0.957✓	0.999✓	0.996✓
TinyStories (123)	0.434✓	0.433✓	0.256✓	0.065✓	0.913×	0.199✓	0.328✓	0.039✓
the_stack (1000)	0✓	4.16e-237✓	1.16e-211✓	2.30e-76✓	0.999✓	0.313×	0.995✓	0.976✓
the_stack (123)	0.999✓	0.993✓	0.610✓	0.491✓	0.916×	0.255✓	0.328✓	0.038✓

of $\alpha = 0.01$. We stress that the absolute magnitude of extremely small p -values is not meaningful: once p falls below the numerical and sampling-noise floor (e.g., $< 10^{-20}$), a value such as 10^{-260} should not be read as stronger evidence than 10^{-20} , since both decisively reject the null.

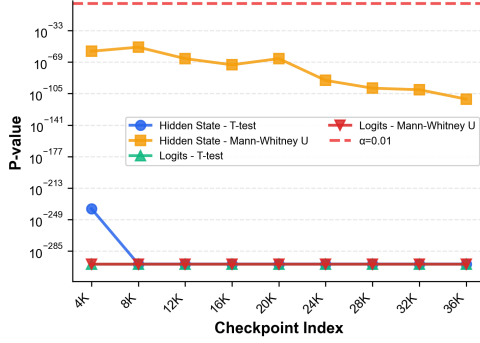
Distinct initialization seeds produce distinct fingerprints. Table 9a and Table 9b report the p -values of our correlation test for pairs of models initialized with different random seeds (42, 123, 1000, and 2000), under both the t -test and the U -test, for the LLaMA-style and Qwen-style models respectively. Every p -value satisfies $p > 0.01$, so the test reliably separates models initialized from different seeds. Distinct seeds therefore yield distinct fingerprint behaviors, permitting models to be separated “at birth.”

Training preserves the initialization fingerprint. Table 10 (LLaMA-style) and Table 11 (Qwen-style) compare each initialization model s^{init} against its descendant s^{base} , trained on the OpenWebText corpus (Gokaslan et al., 2019) ($\approx 10B$ tokens). For every seed–model pair the p -value satisfies $p < 0.01$, indicating that the bias profiles remain strongly correlated and thus share a common lineage; the trained model inherits the fingerprint of its initialization, and training does not erase it. The baseline methods evaluated in Table 10 uniformly fail to discriminate across seeds, which again indicates that their separability originates in training-induced artifacts rather than in initialization.

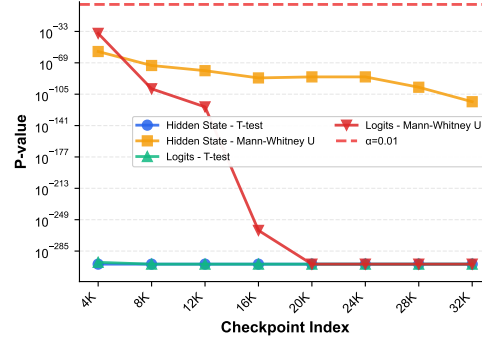
Identical data and ordering do not cause fingerprints to converge. In Table 13a (LLaMA-style) and Table 13b (Qwen-style), all four “suspect” models s_i^{base} with $i \in \{42, 123, 100, 2000\}$ are trained on the identical corpus (OpenWebText) under the identical data ordering; we fix the training seed to lock the ordering, so that the sole point of variation is the initialization seed i . The aim is to test whether fingerprint behavior is erased or confounded by identical data and ordering. Across every

Table 13: An identical corpus and data ordering do not render fingerprint behavior identical across distinct initializations.

(a) LLaMA-style models					(b) Qwen-style models				
Model Pair	Logits Output		Hidden State		Model Pair	Logits Output		Hidden State	
	<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test		<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test
s_{123}^{init} vs. s_{1000}^{base}	0.484	0.500	0.385	0.486	s_{123}^{init} vs. s_{1000}^{base}	0.598	0.638	0.286	0.254
s_{1000}^{init} vs. s_{2000}^{base}	0.946	0.956	0.135	0.096	s_{1000}^{init} vs. s_{123}^{base}	0.804	0.805	0.236	0.240
s_{42}^{init} vs. s_{123}^{base}	0.598	0.589	0.426	0.337	s_{42}^{init} vs. s_{2000}^{base}	0.589	0.608	0.226	0.2043
s_{2000}^{init} vs. s_{42}^{base}	0.756	0.781	0.388	0.287	s_{2000}^{init} vs. s_{42}^{base}	0.523	0.482	0.312	0.323



(a) LLaMA-style models



(b) Qwen-style models

Figure 5: The fingerprint verifies lineage at every checkpoint (p -value < 0.01) for both model families.

cross-seed pair the p -value remains > 0.01 , in sharp contrast to the near-zero values of Table 10. The fingerprint is thus seed-specific even under identical data and curriculum.

Continual training on heterogeneous datasets does not confound the fingerprint. The preceding experiments serve solely to establish that SeedPrints can act as a biometric fingerprint. From a copyright standpoint, the inability of prior methods to distinguish models with different seeds is not itself a deficiency, since initialization seeds carry no well-defined copyright status. The consequential fragility is instead that their attribution is readily misled by the *data distribution, failing to recognize lineage when the training distribution shifts substantially during continued training.*

In Table 12a and Table 12b, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two markedly different corpora: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We contrast (i) true descendants trained from the base against (ii) *distractors* derived from a different base model (initialized with seed 123 and trained with a different data ordering on OpenWebText) and then continued on the *same* corpus. The question is whether an attribution method can identify which descendant genuinely shares lineage with the base.

We observe that every prior baseline fails under the code setting (The Stack), misclassifying true descendants as distractors. This indicates that these methods primarily track domain similarity rather than lineage identity: TinyStories is distributionally closer to the pretraining corpus (OpenWebText), whereas The Stack diverges sharply, and a distribution shift of this magnitude readily evades detection. Our method, in contrast, correctly attributes lineage under both corpora. The proposed fingerprint is therefore not a proxy for the data distribution: it withstands substantial domain shift and persists beyond the initial pretraining stage.

D.2.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. We first conduct a controlled pretraining experiment on OpenWebText, in which the initialization seed is known. Treating each intermediate checkpoint as the base, we test whether our method reliably identifies its offspring along the same training trajectory.

Table 14: Cross-size similarity between Qwen-2.5-14B (target) and the remaining Qwen-2.5 models.

	Qwen-2.5-0.5B	Qwen-2.5-1.5B	Qwen-2.5-3B	Qwen-2.5-7B
SeedPrint ($\uparrow$)	0.7184	0.9707	0.3898	0.5429
REEF ($\downarrow$)	0.8250	0.8399	0.8833	0.8461
HUREF ($\downarrow$)	-6.37×10^{-5}	-2.84×10^{-5}	-8.70×10^{-6}	-6.08×10^{-6}

Every variant consistently recognizes the suspect as belonging to the same lineage, with p -values remaining below the 0.01 threshold (Figure 5a for the LLaMA-style models and Figure 5b for the Qwen-style models). This confirms the stability of SeedPrints under continuous optimization.

D.3 CROSS-SIZE FINGERPRINT WITHIN THE QWEN-2.5 SERIES

To examine whether SeedPrints “collapses” within a model family as the parameter count varies, we report the cross-size similarity between Qwen-2.5-14B (target) and the smaller models of the Qwen-2.5 family. Table 14 presents the results for SeedPrint ($\uparrow$) alongside two baselines, REEF ($\downarrow$) and HUREF ($\downarrow$).

To probe whether initialization-born fingerprints “collapse” onto a family-level signature under model-size scaling, we conduct a cross-size study within the Qwen-2.5 series. We designate Qwen-2.5-14B as the protected target and treat Qwen-2.5-{0.5B, 1.5B, 3B, 7B} as suspects that share the architectural family but are not trained as descendants of Qwen-2.5-14B. As discussed in the main paper, SeedPrints returns a p -value for the hypothesis “the suspect shares initialization-born fingerprints with the target,” whereas REEF and HUREF are similarity-based baselines that emit continuous scores and require the user to choose a threshold.

Several observations follow. First, every SeedPrints p -value lies comfortably above a typical significance level (e.g., $\alpha=0.01$). Under the hypothesis-testing interpretation, none of the smaller Qwen-2.5 models is therefore flagged as sharing the lineage of Qwen-2.5-14B. This is the desired behavior: although these models belong to the same family and reuse a similar architecture, the fingerprinting test does not collapse onto a single family-level signature that would spuriously conflate distinct pretraining runs.

The cross-size p -values populate different regions of $[0, 1]$ rather than concentrating at a single value. We do *not* interpret these magnitudes as a continuous “similarity score”; their role is to support a calibrated binary decision at a fixed significance level. The observed spread nonetheless indicates that the test statistic remains sensitive to the specific combination of architecture and initialization seed rather than degenerating into a generic pattern shared by all Qwen-2.5 variants.

Finally, the similarity-based baselines behave differently. REEF produces scores confined to a narrow band (0.8250–0.8833) across all sizes, whereas HUREF produces values numerically close to zero (of order 10^{-5}). Because neither method supplies a built-in decision rule, an operator must set thresholds manually, and it is unclear how a single cutoff could robustly separate cross-size variants from genuinely lineage-related models. SeedPrints, in contrast, directly yields a statistical decision at a prescribed significance level, providing a clearer and more operational answer to the question “do these two models plausibly share a training lineage?” within a model family.

E MODEL CATALOG AND DECODING SETTINGS

Our evaluation covers 58 models in total. Below we document the decoding configuration applied uniformly across all runs and enumerate the model catalog grouped by family. For each derived model we additionally specify the post-training transformation (*finetune*, *adapter*, *merge*, *quantization*, or *distillation*). “PT” and “IT” denote the *pre-trained* and *instruction-tuned* source models, respectively.

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Parameter	Value
max_new_tokens	512
temperature	0.7
top_p	0.9
top_k	50
do_sample	true
max_input_length	512

E.1 MODEL LIST BY FAMILY

E.1.1 QWEN-2.5-7B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-7B	—
IT	Qwen/Qwen2.5-7B-Instruct	—
Derived	Qwen/Qwen2.5-Math-7B	finetune
Derived	Qwen/Qwen2.5-Coder-7B-Instruct	finetune
Derived	WangCa/Qwen2.5-7B-Medicine	finetune
Derived	huihui-ai/Qwen2.5-7B-Instruct-abliterated-v2	finetune
Derived	Locutusque/StockQwen-2.5-7B	merge
Derived	bunnycore/QevaCoT-7B-Stock	merge
Derived	fangcaotank/task-10-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	SeeFlock/task-12-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4	quantization
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8	quantization
Derived	Lansechen/Qwen2.5-7B-Open-R1-Distill	distillation

E.1.2 QWEN2.5-14B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-14B	—
IT	Qwen/Qwen2.5-14B-Instruct	—
Derived	Qwen/Qwen2.5-Coder-14B	finetune
Derived	oxyapi/oxy-1-small	finetune
Derived	v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	merge
Derived	ToastyPigeon/qwen-story-test-qlora	adapter
Derived	Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4	quantization
Derived	deepseek-ai/DeepSeek-R1-Distill-Qwen-14B	distillation

E.1.3 LLAMA-3.1-8B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-3.1-8B	—
IT	meta-llama/Llama-3.1-8B-Instruct	—
Derived	ValiantLabs/Llama3.1-8B-Fireplace2	finetune
Derived	RedHatAI/Llama-3.1-8B-tldr	finetune
Derived	proxectonos/Llama-3.1-Carballo	finetune
Derived	mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated	finetune
Derived	gaverfraxz/Meta-Llama-3.1-8B-Instruct-HalfAbliterated-TIES	merge
Derived	Xiaojian9992024/Llama3.1-8B-ExtraMix	merge
Derived	LlamaFactoryAI/Llama-3.1-8B-Instruct-cv-job-description-matching	adapter
Derived	chchen/Llama-3.1-8B-Instruct-PsyCourse-fold7	adapter
Derived	iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8	quantization
Derived	DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ	quantization
Derived	asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill	distillation

E.1.4 MISTRAL-7B-v0.3

Variant	HuggingFace repo	Type
PT	mistralai/Mistral-7B-v0.3	—
IT	mistralai/Mistral-7B-Instruct-v0.3	—
Derived	KurmaAI/AQUA-7B	finetune
Derived	openfoodfacts/spellcheck-mistral-7b	finetune
Derived	grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B	merge
Derived	chaymaemrhrioui/mistral-Brain_Model_ACC_Trainer	adapter
Derived	RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit	quantization
Derived	eganwo/mistral7b-distilled-from-deepseek-r1-qwen32b	distillation

E.1.5 GEMMA-2-2B

Variant	HuggingFace repo	Type
PT	google/gemma-2-2b	—
IT	google/gemma-2-2b-it	—
Derived	rinna/gemma-2-baku-2b	finetune
Derived	anakin87/gemma-2-2b-neogenesis-ita	finetune
Derived	vonjack/gemma2-2b-merged	merge
Derived	google-cloud-partnership/gemma-2-2b-it-lora-sql	adapter
Derived	qilowoq/gemma-2-2b-it-4Bit-GPTQ	quantization
Derived	Syed-Hasan-8503/Gemma-2-2b-it-distilled	distillation

E.1.6 LLAMA-2-7B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-2-7b-hf	—
IT	meta-llama/Llama-2-7b-chat-hf	—
Derived	allenai/tulu-2-7b	finetune
Derived	QIAIUNCC/EYE-Llama_qa	finetune
Derived	DevQuasar/coma-7B-v0.1	merge
Derived	Ammar-1/llama2-Better-Tune	adapter
Derived	TheBloke/Llama-2-7B-Chat-GPTQ	quantization
Derived	cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	distillation

E.1.7 TINYLLAMA-1.1B

Variant	HuggingFace repo	Type
PT/IT	TinyLlama/TinyLlama-1.1B-Chat-v1.0	—
Derived	alexredna/TinyLlama-1.1B-Chat-v1.0-reasoning-v2	finetune
Derived	Edentns/DataVortexTL-1.1B-v0.1	finetune
Derived	appvoid/dot-v2.7	merge
Derived	barissglc/tinyllama-tarot-v1	adapter
Derived	TheBloke/TinyLlama-1.1B-Chat-v1.0-GPTQ	quantization
Derived	anudaw/distilled-code-llama	distillation

Notes. “Type” denotes the post-training transformation applied relative to the PT/IT source model: *finetune* comprises supervised and preference-optimization variants; *adapter* comprises LoRA/QLoRA-style modules; *merge* comprises model soups and TIES-style merges; *quantization* comprises GPTQ/INTx variants; and *distillation* comprises student models distilled from reasoning-augmented teachers.

E.2 SAMPLE RESULTS

We present partial results from a single LeafBench evaluation run in Table 15 (u -test with the simulation-based null). The reported metric is $1 - p$, treated as a similarity score. We adopt a significance level of 0.01 for the p -values; empirically, LeafBench identifies an optimal decision threshold of 0.9920, further corroborating the reliability of our method.

Crucially, although individual p -values should not be read as linear similarity measures, their distribution across model pairs is informative: same-lineage pairs yield substantially smaller p -values, whereas unrelated pairs remain approximately uniform.

Model	Qwen-2.5-7B	Qwen-2.5-7B-Instruct	Qwen2.5-14B	Qwen2.5-14B-Instruct
Qwen-2.5-7B	1.000	1.000	0.079	0.000
Qwen-2.5-7B-Instruct	1.000	1.000	0.813	0.536
Qwen2.5-7B-Math	1.000	0.994	0.047	0.987
Qwen2.5-7B-Coder	0.996	0.999	0.177	0.002
Qwen2.5-7B-Instruct-Medicine	1.000	1.000	0.576	0.369
Qwen2.5-7B-Instruct-Abiliterated	1.000	1.000	0.276	0.886
Qwen2.5-7B-Stock	1.000	1.000	0.939	0.225
QevaCoT-7B	1.000	1.000	0.199	0.119
Qwen2.5-7B-Instruct-Task-10	1.000	1.000	0.672	0.090
Qwen2.5-7B-Instruct-Task-12	1.000	1.000	0.155	0.987
Qwen2.5-7B-Instruct-Int4	1.000	1.000	0.827	0.961
Qwen2.5-7B-Instruct-Int8	1.000	1.000	0.914	0.739
Qwen2.5-7B-Open-R1-Distill	1.000	1.000	0.302	0.955
Qwen2.5-14B	0.324	0.487	1.000	1.000
Qwen2.5-14B-Instruct	0.403	0.702	1.000	1.000
Qwen2.5-Coder-14B	0.492	0.660	1.000	1.000
oxy-1-small	0.492	0.104	1.000	1.000
Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	0.856	0.931	1.000	1.000
Qwen-story-test-qlora	0.112	0.898	1.000	1.000
Qwen2.5-14B-Instruct-GPTQ-Int4	0.583	0.092	1.000	1.000
DeepSeek-R1-Distill-Qwen-14B	0.204	0.000	1.000	1.000
Llama-3.1-8B	0.638	0.714	0.156	0.000
Llama-3.1-8B-Instruct	0.643	0.118	0.128	0.510
Llama-3.1-8B-Fireplace2	0.715	0.532	0.065	0.354
Llama-3.1-8B-TLDR	0.925	0.811	0.331	0.014
Llama-3.1-8B-Carballo	0.001	0.605	0.003	0.946
Llama-3.1-8B-Instruct-Abiliterated	0.061	0.005	0.121	0.422
Llama-3.1-8B-Instruct-HalfAbiliterated-TIES	0.356	0.509	0.796	0.000
Llama-3.1-8B-ExtraMix	0.640	0.160	0.721	0.875
Llama-3.1-8B-Instruct-cv-job-description-matching	0.684	0.292	0.004	0.001
Llama-3.1-8B-Instruct-PsyCourse-fold7	0.682	0.057	0.063	0.191
Llama-3.1-8B-Instruct-8bit	0.798	0.214	0.154	0.385
Llama-3.1-8B-Instruct-4bit	0.513	0.023	0.005	0.933
Llama-3.1-8B-Instruct-Open-R1-Distill	0.800	0.101	0.772	0.024
Mistral-7B-v0.3	0.224	0.352	0.506	0.044
Mistral-7B-v0.3-Instruct	0.000	0.000	0.984	0.104
AQUA-7B	0.014	0.613	0.703	0.569
Mistral-7B-v0.3-Spellcheck	0.005	0.481	0.872	0.105
Mistral-7B-v0.3-Instruct-demi-merge	0.904	0.774	0.058	0.522
Mistral-7B-v0.3-Brain	0.340	0.983	0.481	0.016
Mistral-7B-v0.3-Instruct-GPTQ-4bit	0.002	0.281	0.089	0.784
Mistral-7B-distilled-from-deepseek-r1-qwen32b	0.009	0.001	0.511	0.015
Gemma-2-2b	0.747	0.869	0.689	0.000
Gemma-2-2b-it	0.944	0.039	0.626	0.060
Gemma-2-baku-2b	0.616	0.231	0.176	0.464
Gemma-2-2b-neogenesis-ita	0.002	0.536	0.079	0.980
Gemma-2-2b-merged	0.951	0.012	0.651	0.075
Gemma-2-2b-it-lora-sql	0.236	0.408	0.000	0.000
Gemma-2-2B-it-4Bit-GPTQ	0.945	0.159	0.000	0.008
Gemma-2-2b-it-distilled	0.482	0.573	0.271	0.276
Llama-2-7b	0.443	0.470	0.148	0.276
Llama-2-7b-chat	0.528	0.868	0.424	0.766
tulu-2-7b	0.125	0.272	0.170	0.888
EYE-Llama_qa	0.657	0.244	0.674	0.968
coma-7B-v0.1	0.953	0.004	0.573	0.731
llama2-Better-Tune	0.136	0.699	0.469	0.860
Llama-2-7B-Chat-GPTQ	0.594	0.603	0.582	0.424
llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	0.349	0.224	0.366	0.976

Table 15: **Sample results.** Pairwise similarity between target models (rows) and base models (columns), reported as $1 - p$. Cells satisfying $1 - p \geq 0.99$ are highlighted in green and identified as same-lineage pairs.